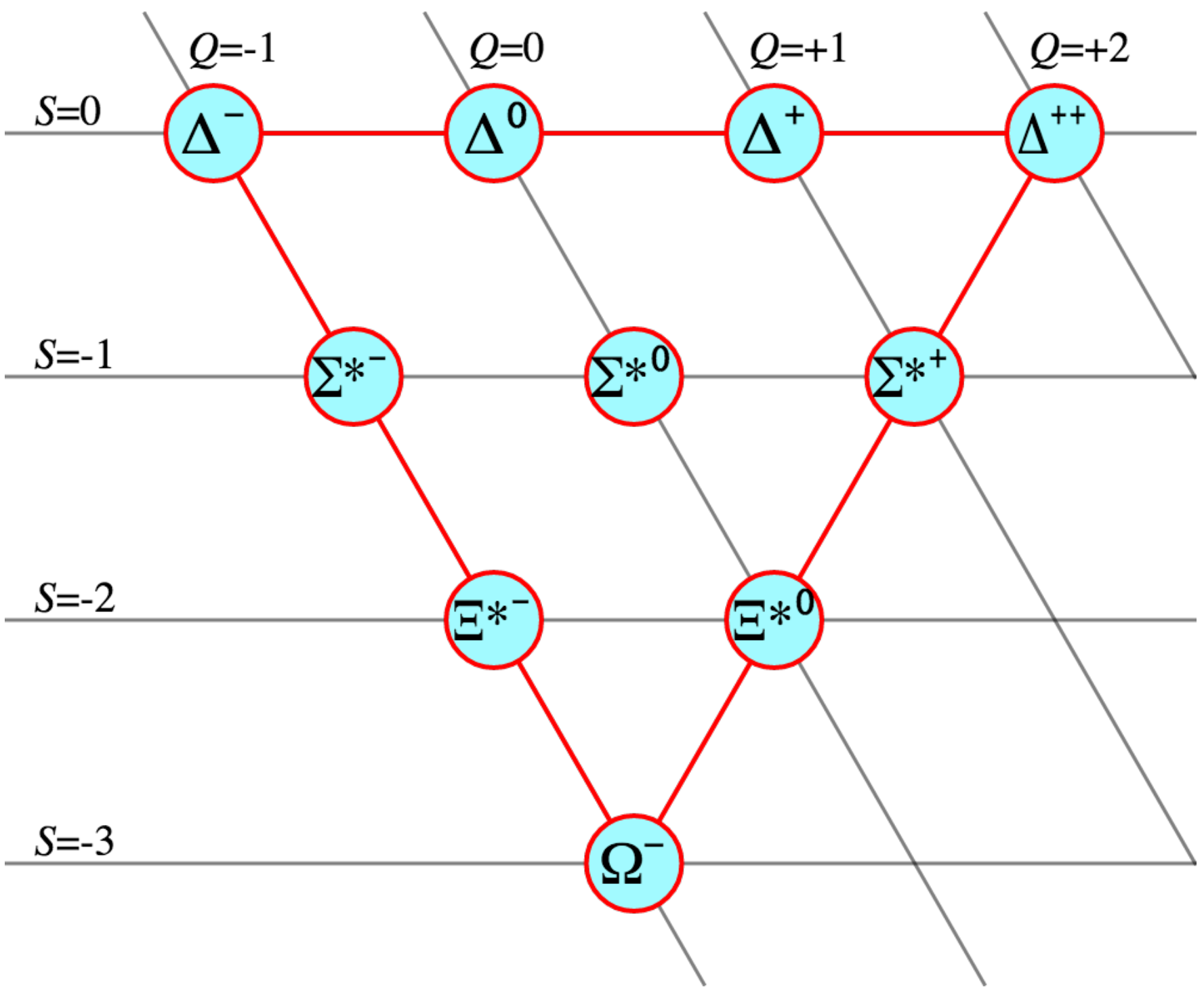
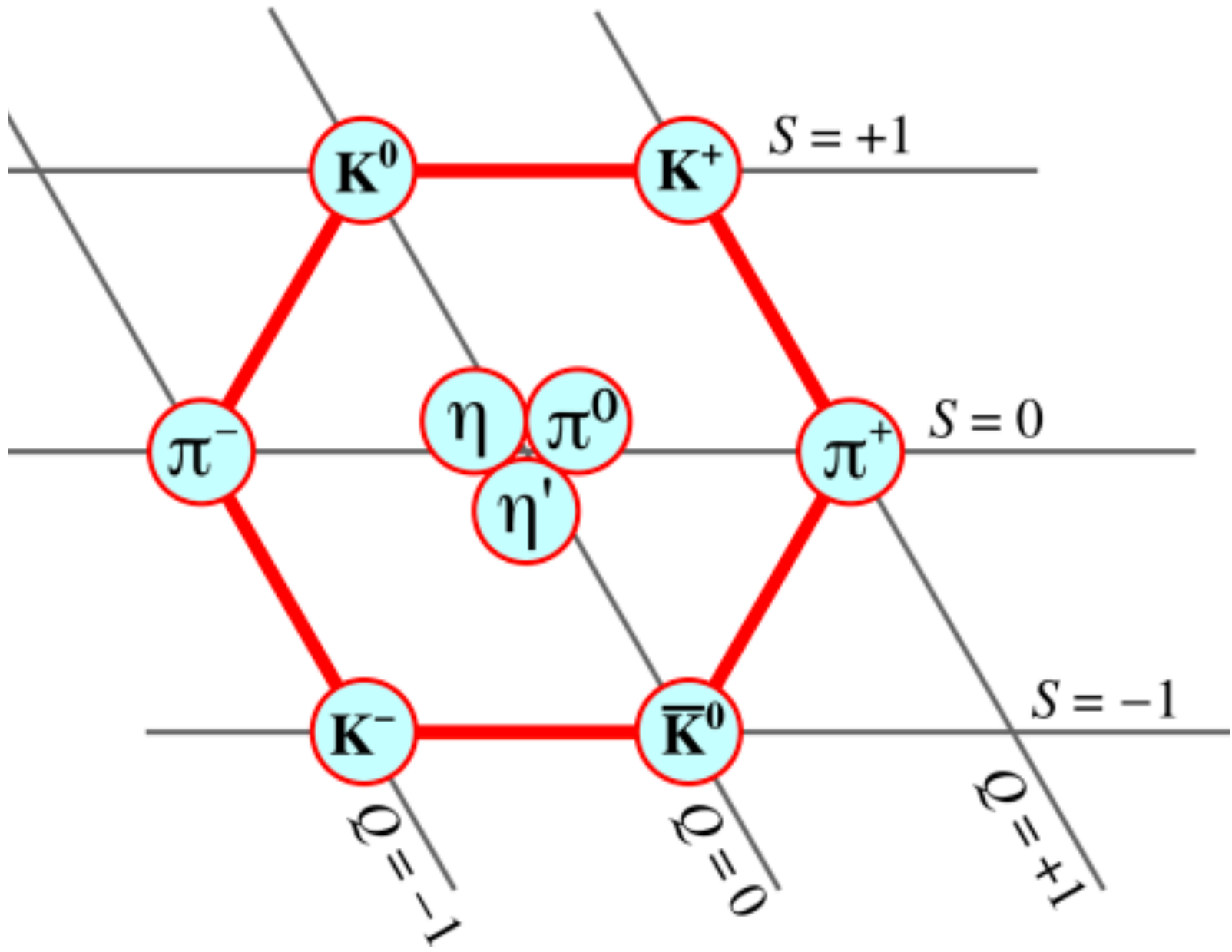
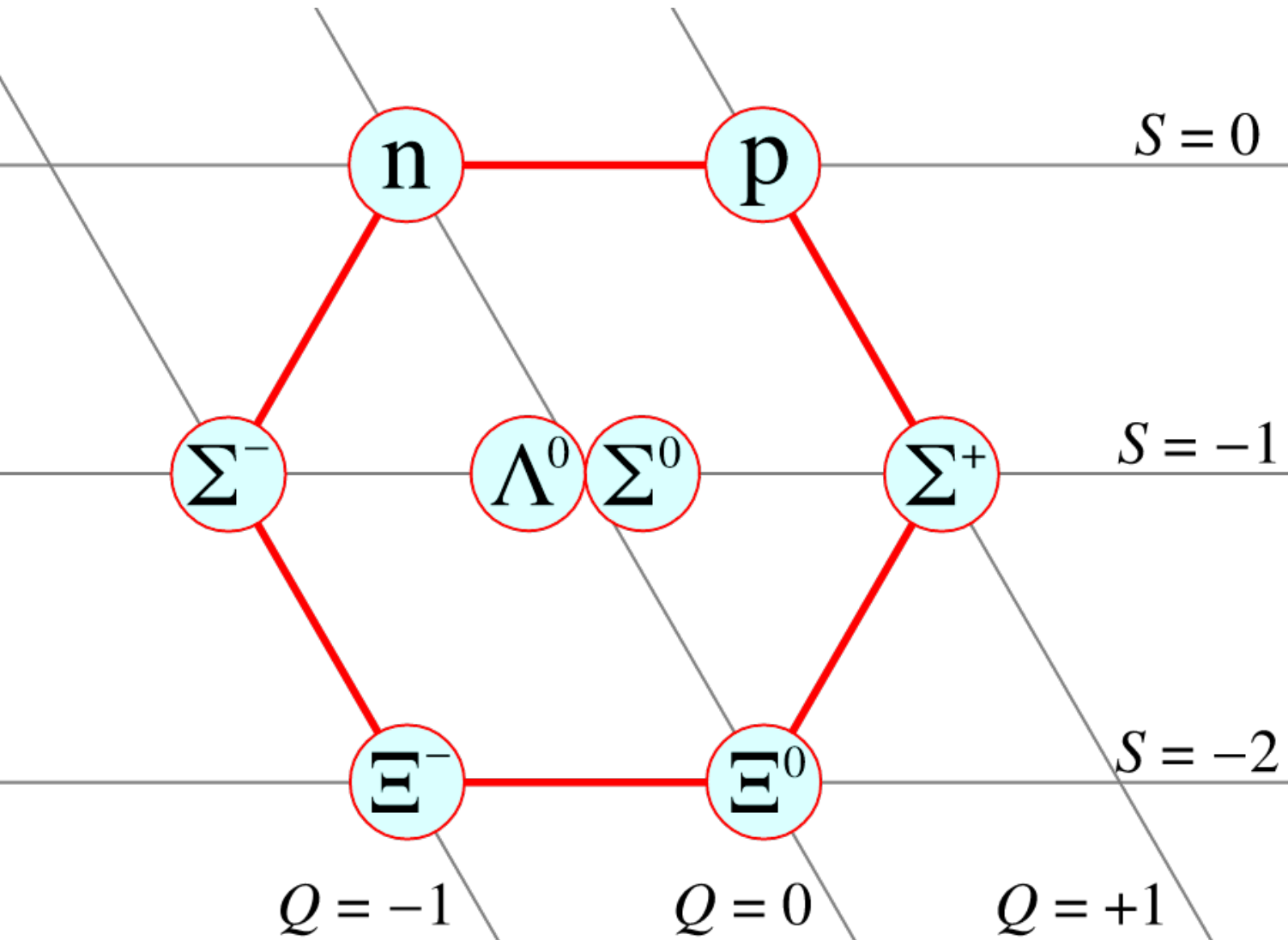
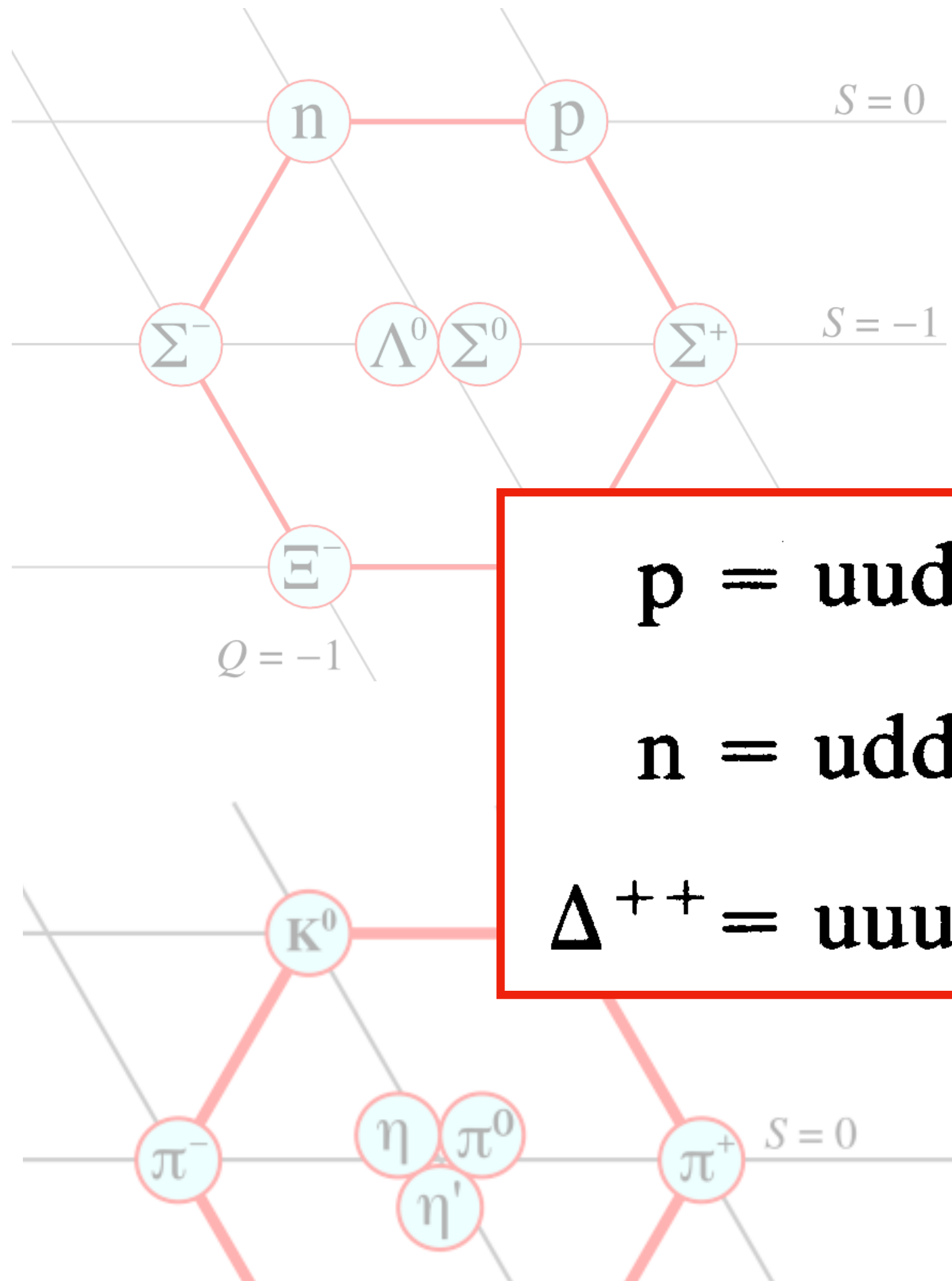


# Introduction to Particle Physics

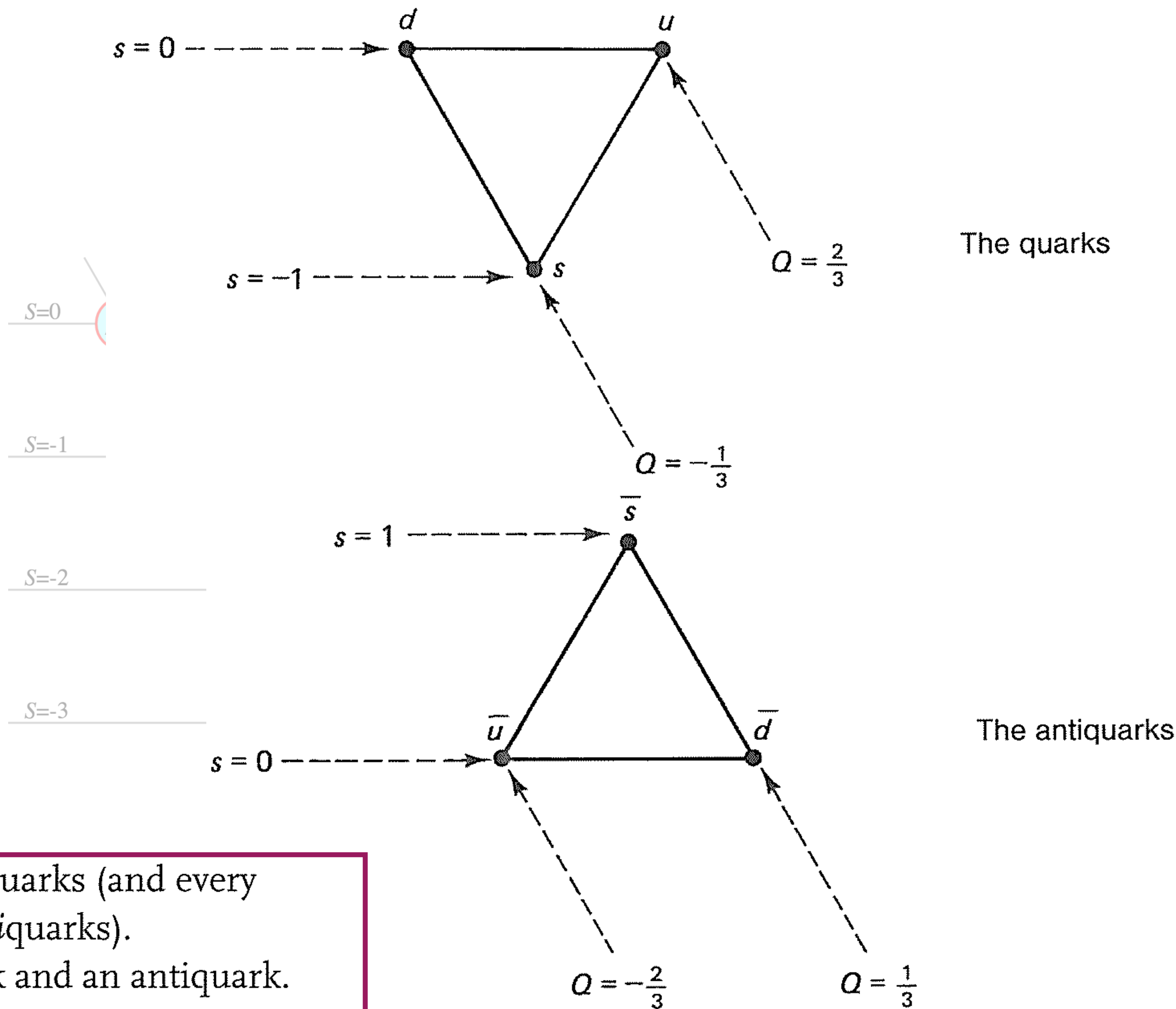
# The Eightfold Way



# The Quark Model



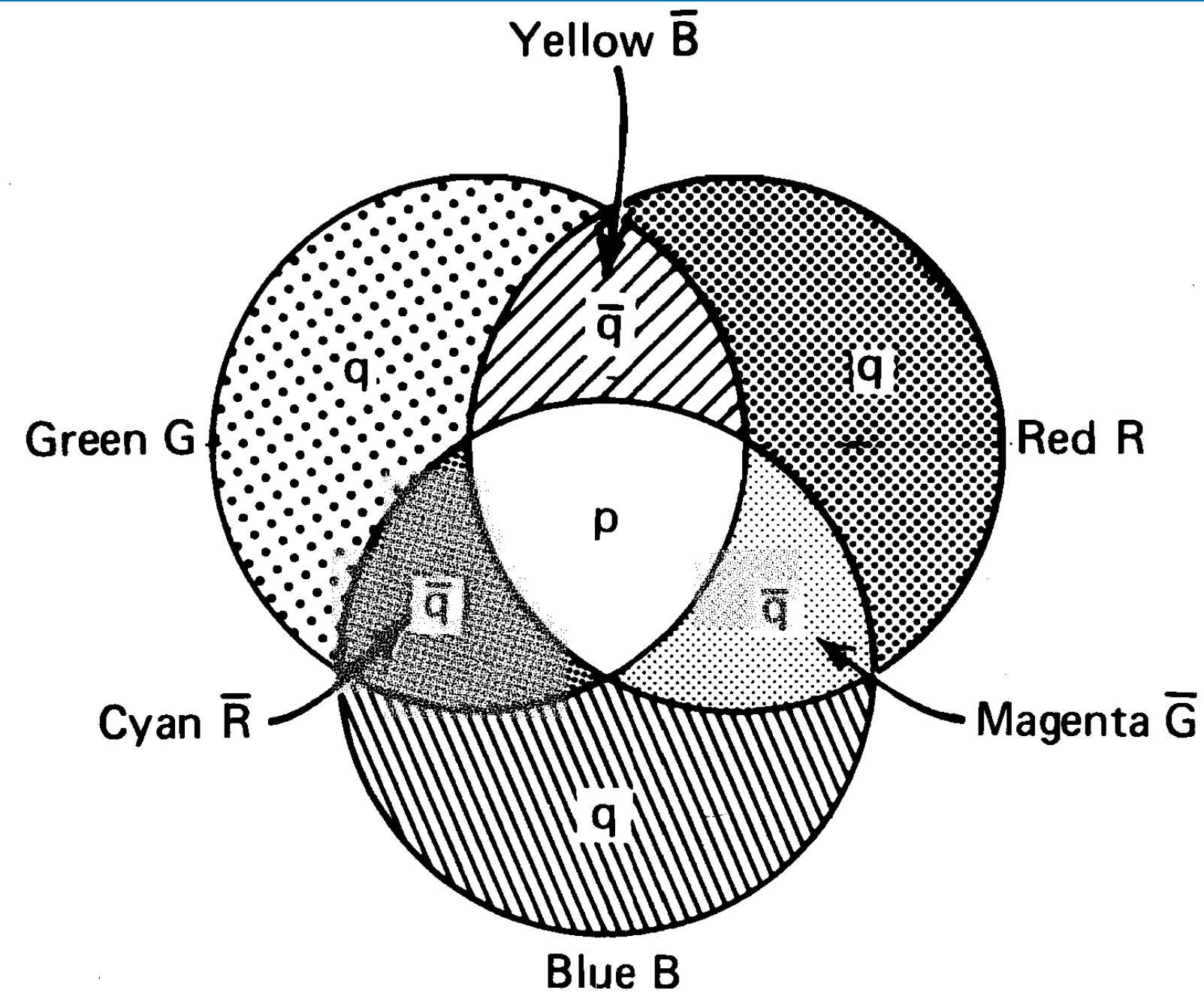
$p = uud$   
 $n = udd$   
 $\Delta^{++} = uuu.$



1. Every baryon is composed of three quarks (and every *antibaryon* is composed of three *antiquarks*).
2. Every meson is composed of a quark and an antiquark.



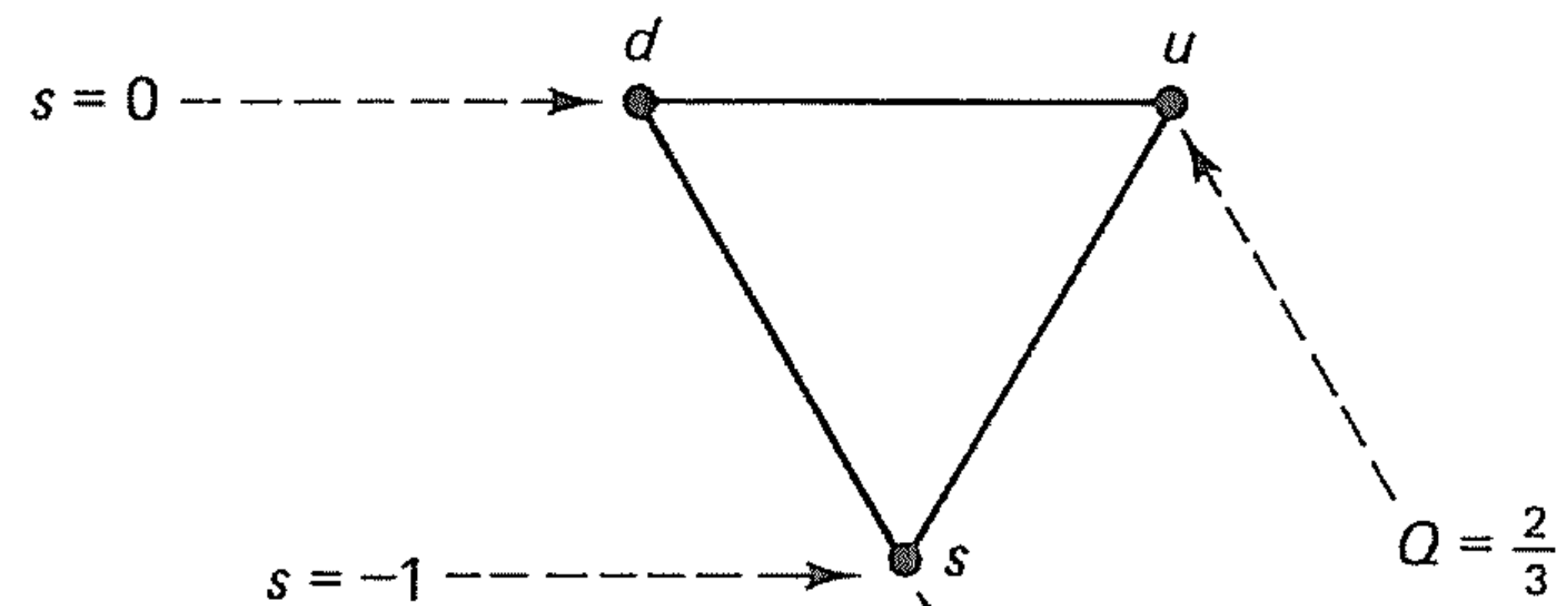
The quarks have colour  
but the "world" is colourless



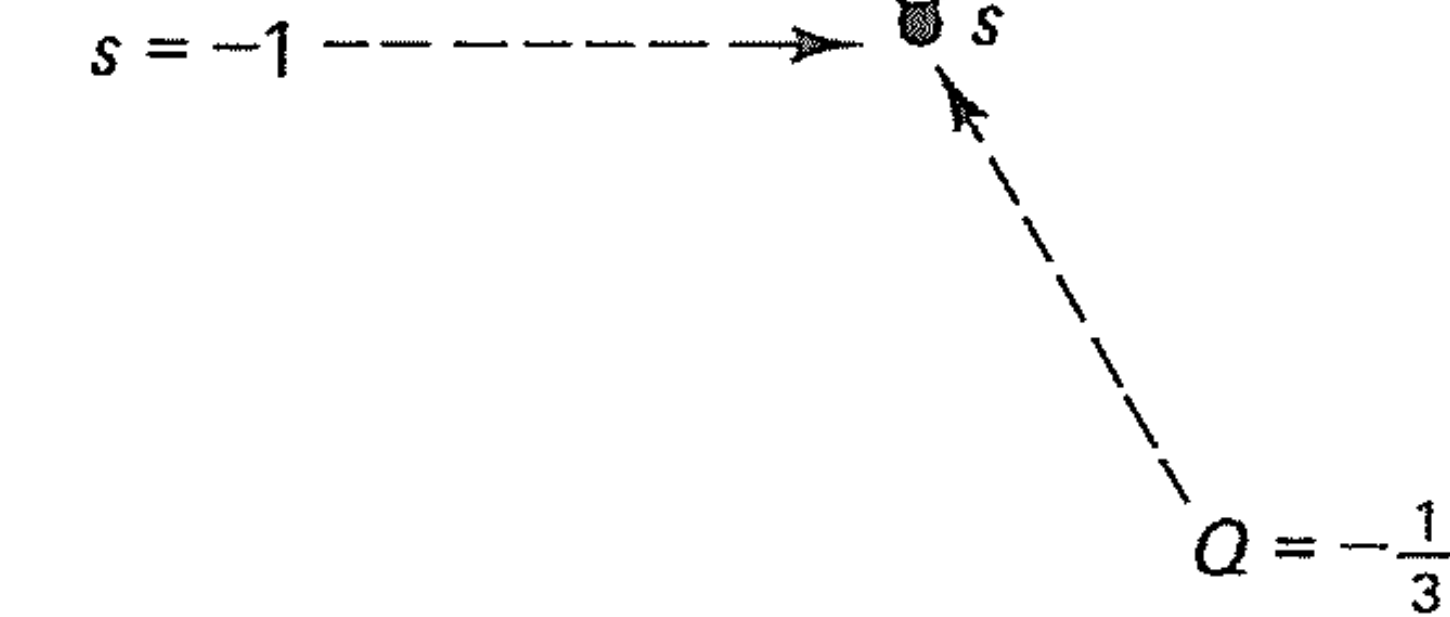
$p = "RGB"$

$\bar{p} = "\bar{R}\bar{G}\bar{B}"$

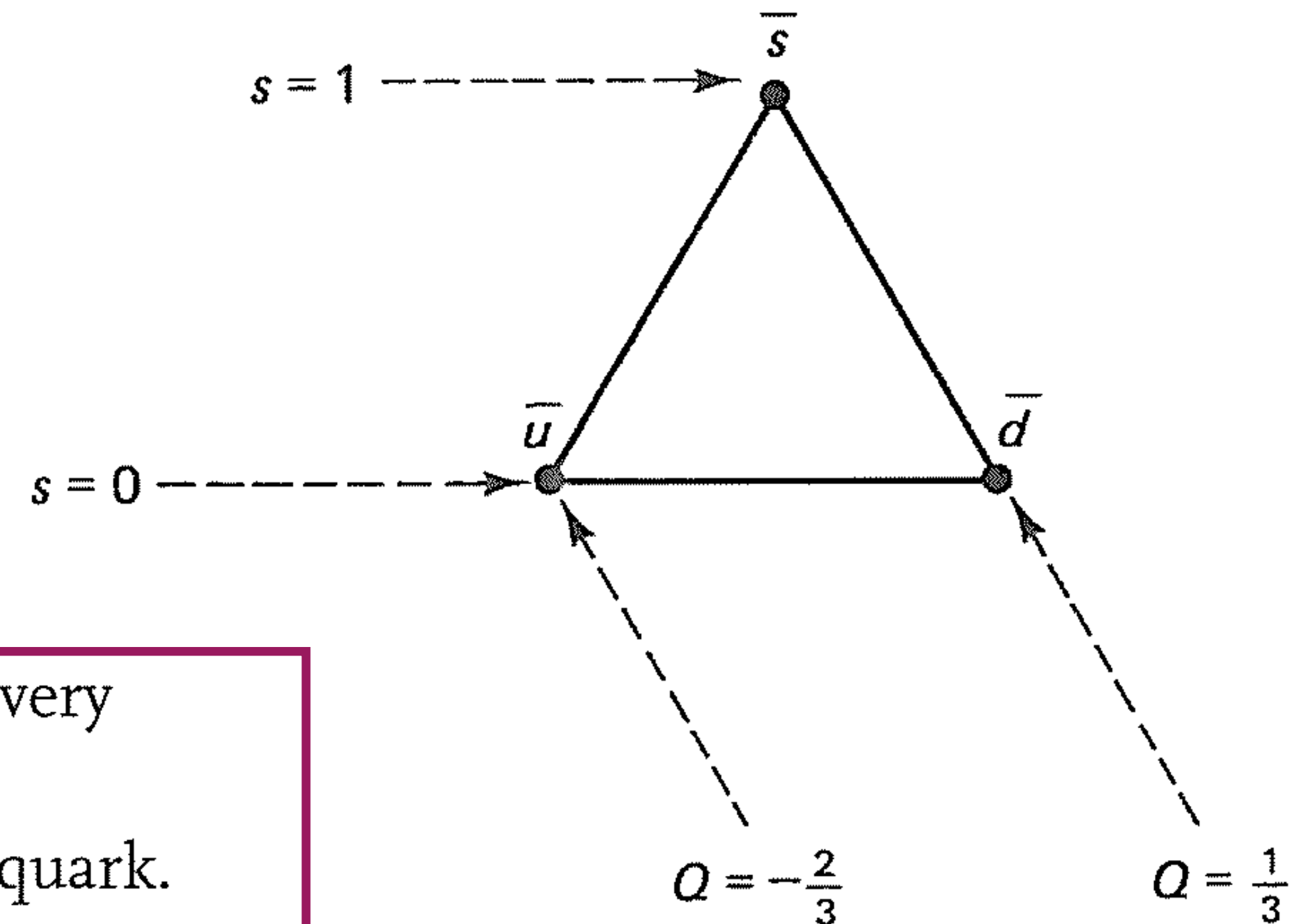
$\pi = "R\bar{R} + B\bar{B} + G\bar{G}"$



The quarks



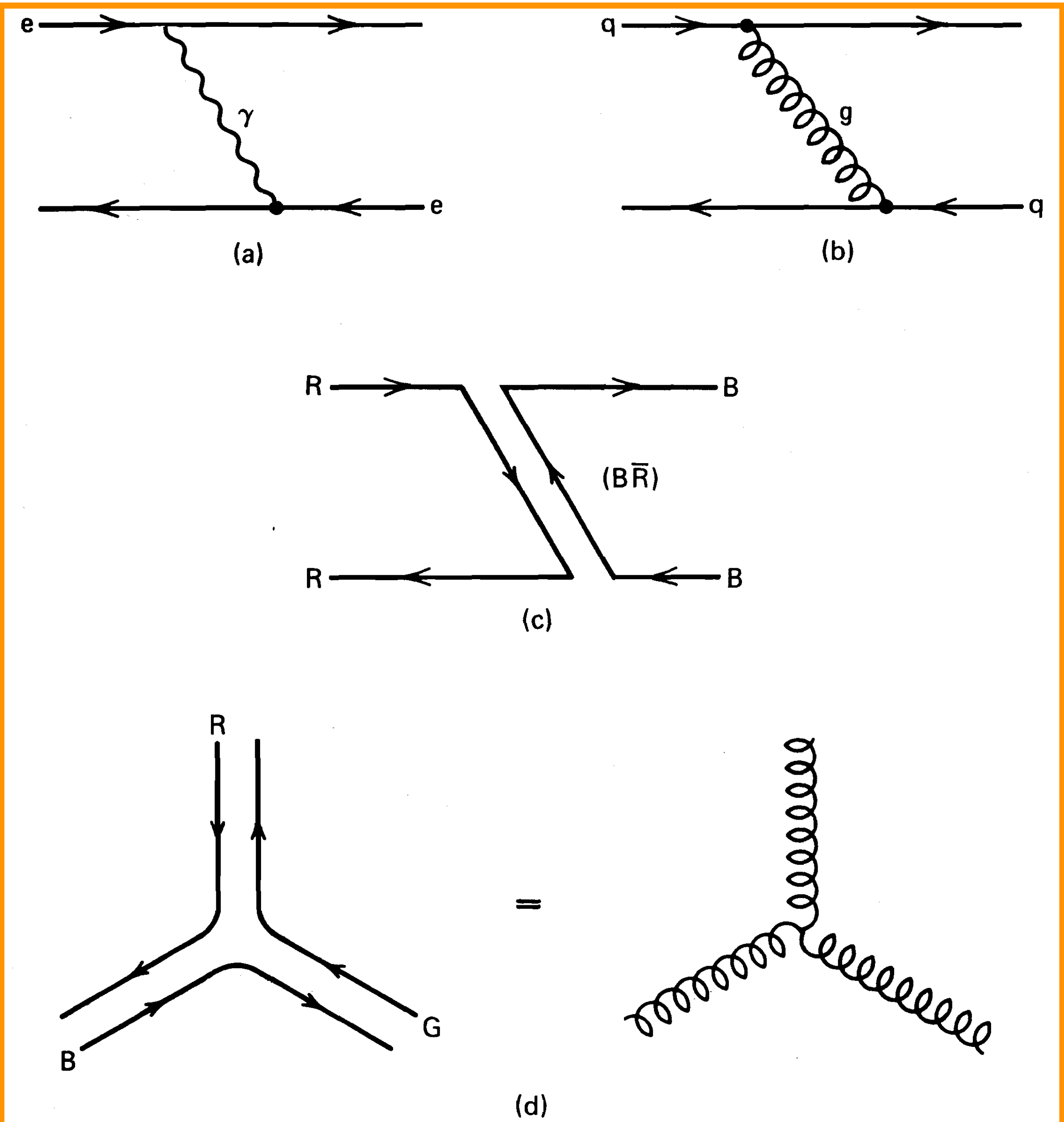
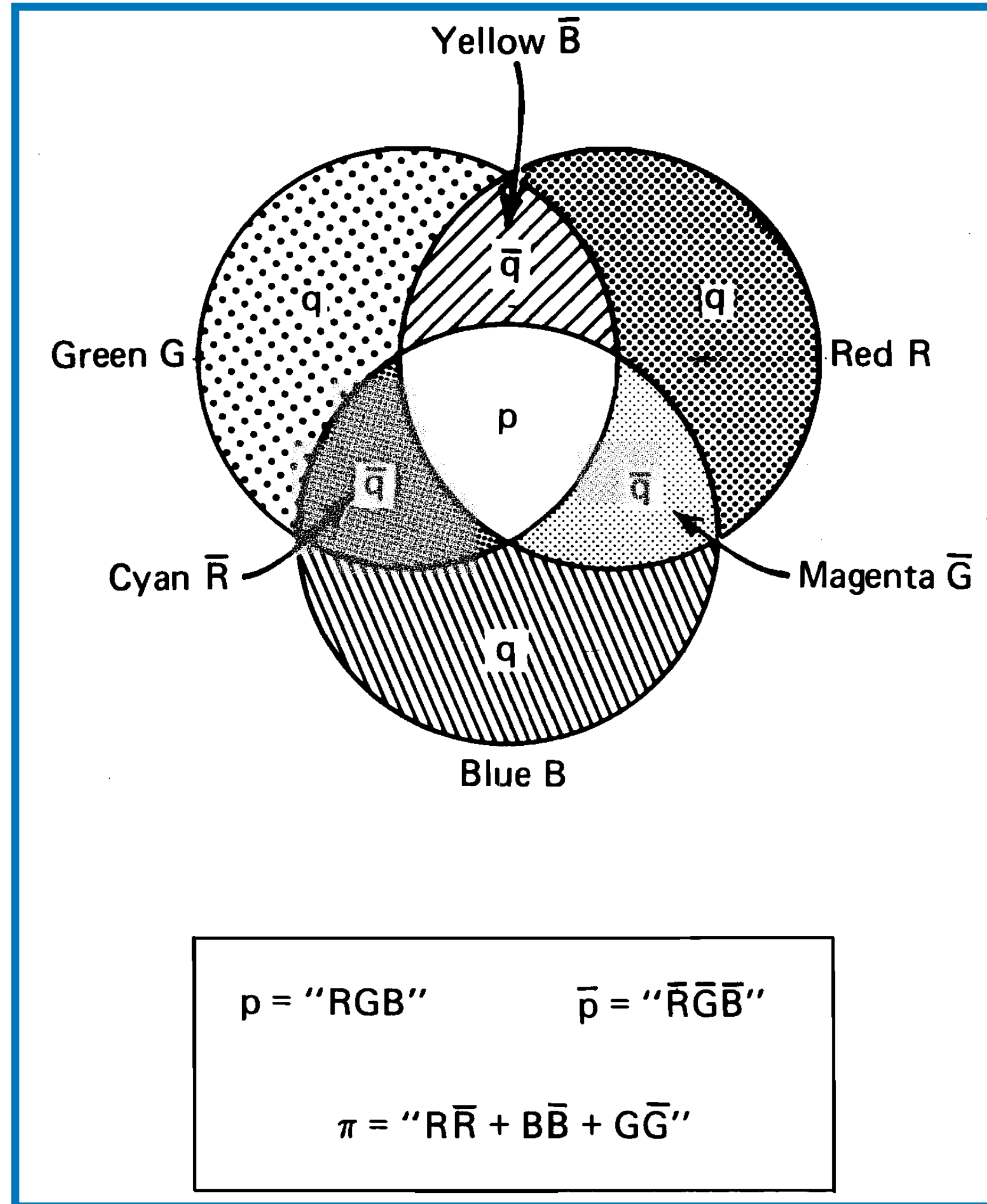
The antiquarks



1. Every baryon is composed of three quarks (and every *antibaryon* is composed of three *antiquarks*).
2. Every meson is composed of a quark and an antiquark.



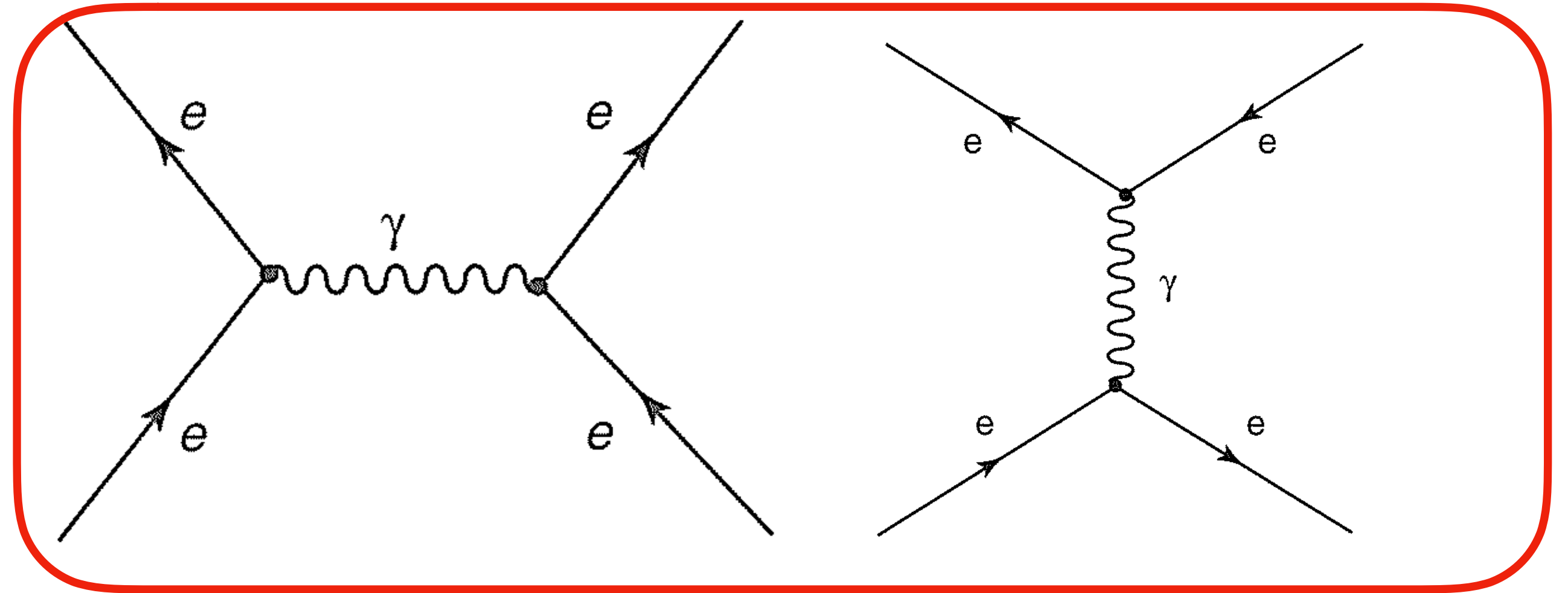
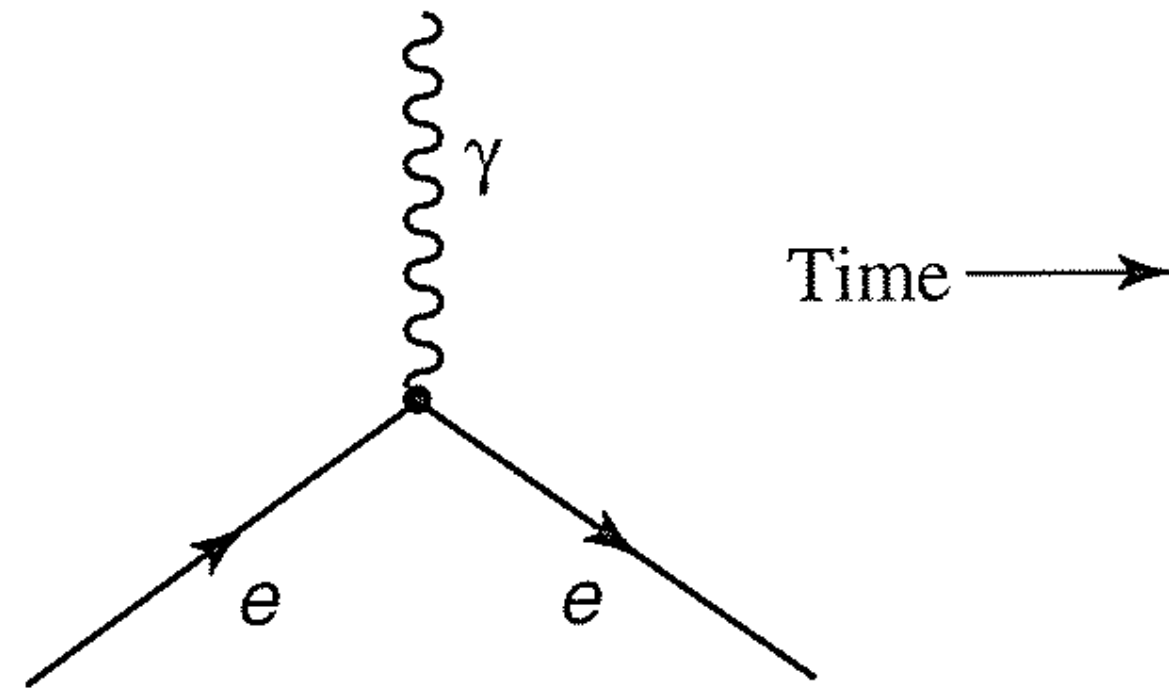
The quarks have colour  
but the “world” is colourless



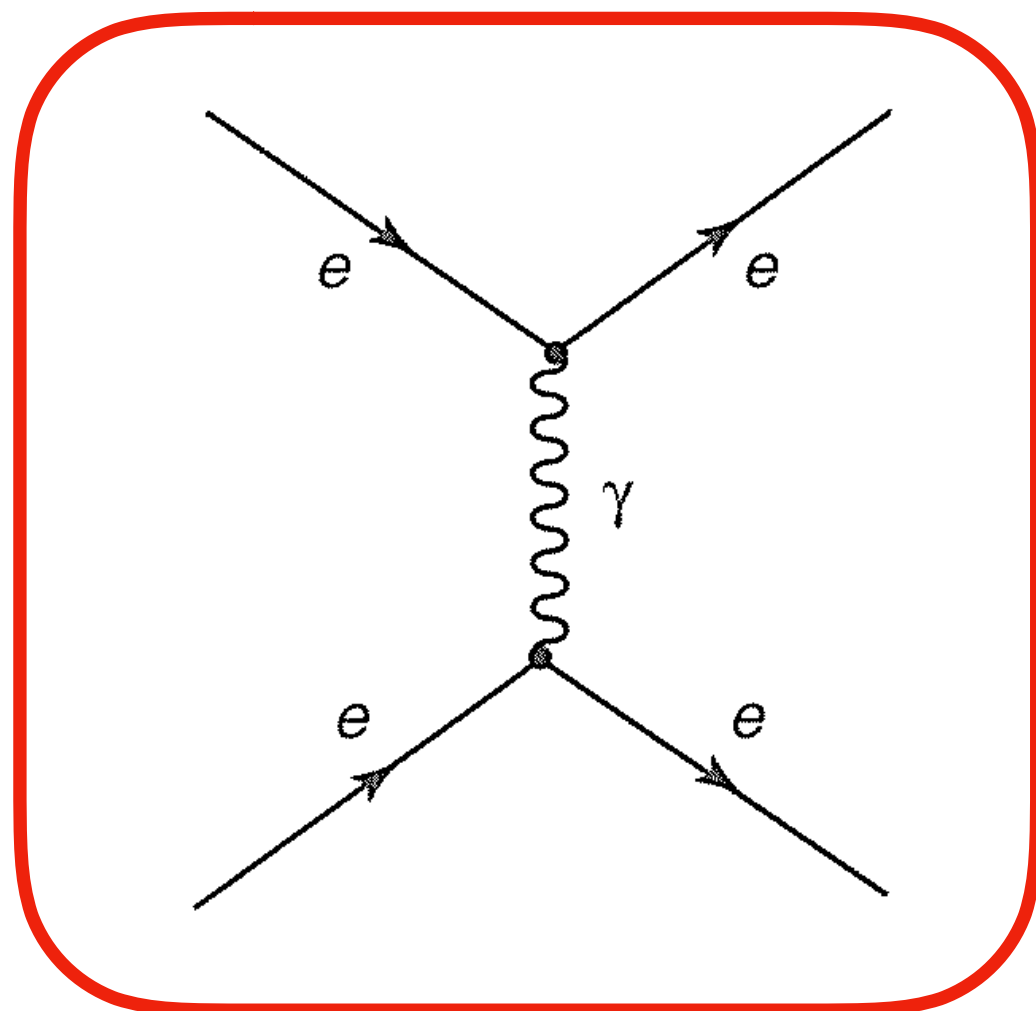
**Fig. 1.4** (a) Electromagnetic interaction by photon exchange. (b) Strong interaction by gluon exchange. (c) Flow of color in (b). (d) Self-coupling of gluons.

# Quantum Electrodynamics

**Bhabha scattering**  
 $e^- + e^+ \rightarrow e^- + e^+$

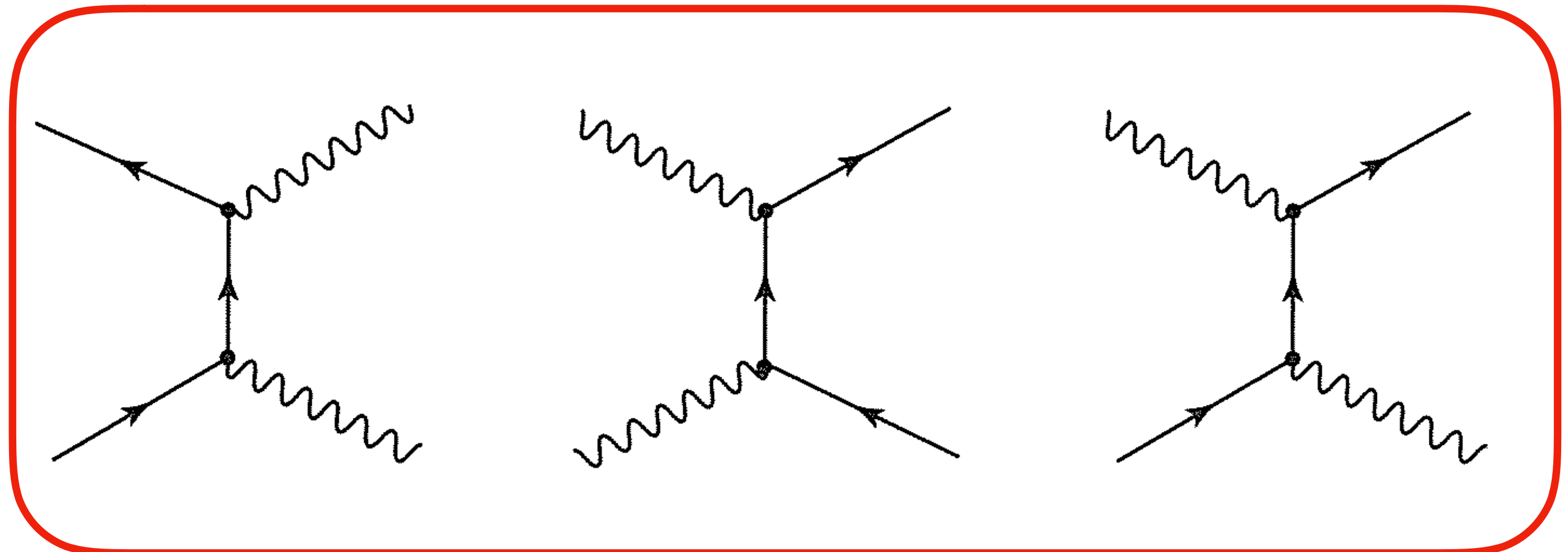


**Møller scattering**  
 $e^- + e^- \rightarrow e^- + e^-$

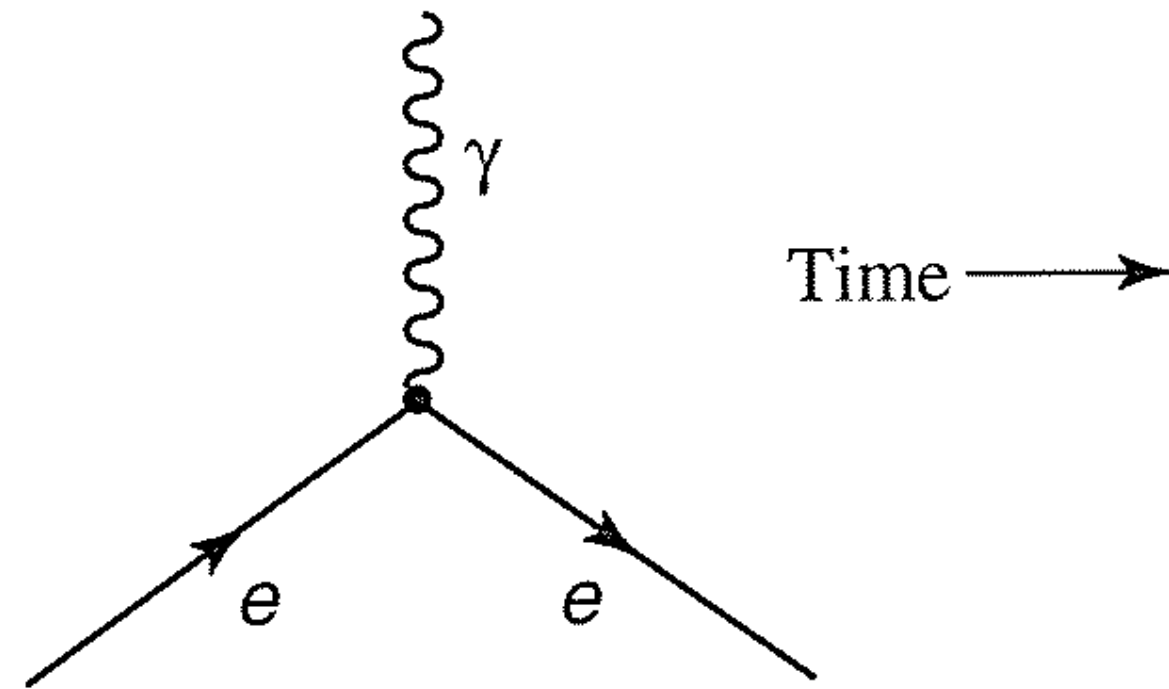


$$A + B \rightarrow C + D, A + \bar{C} \rightarrow \bar{B} + D, A + \bar{D} \rightarrow C + \bar{B}, \bar{C} + \bar{D} \rightarrow \bar{A} + \bar{B}$$

**Compton scattering**



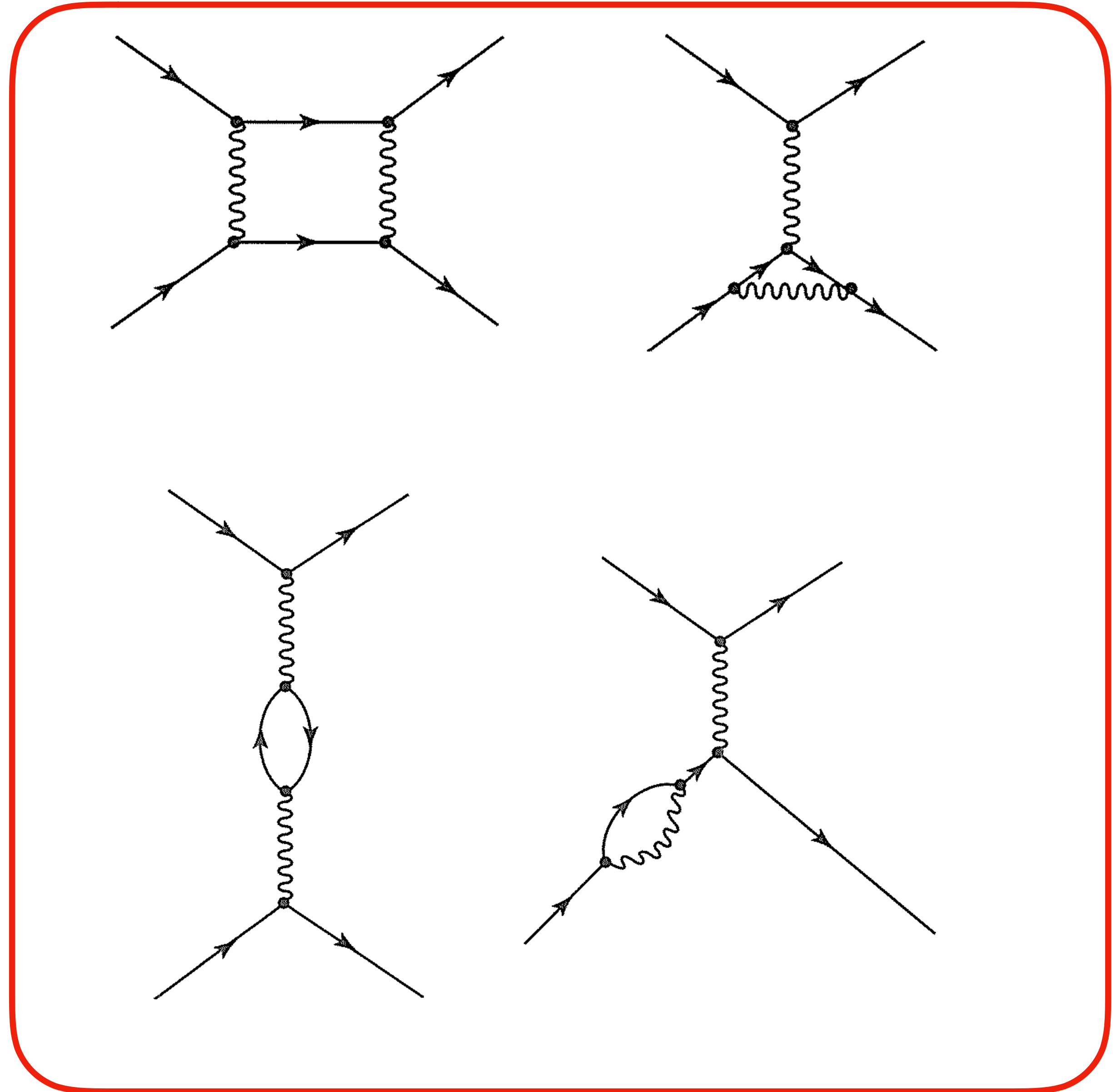
# Quantum Electrodynamics



**Nothing is not empty!**

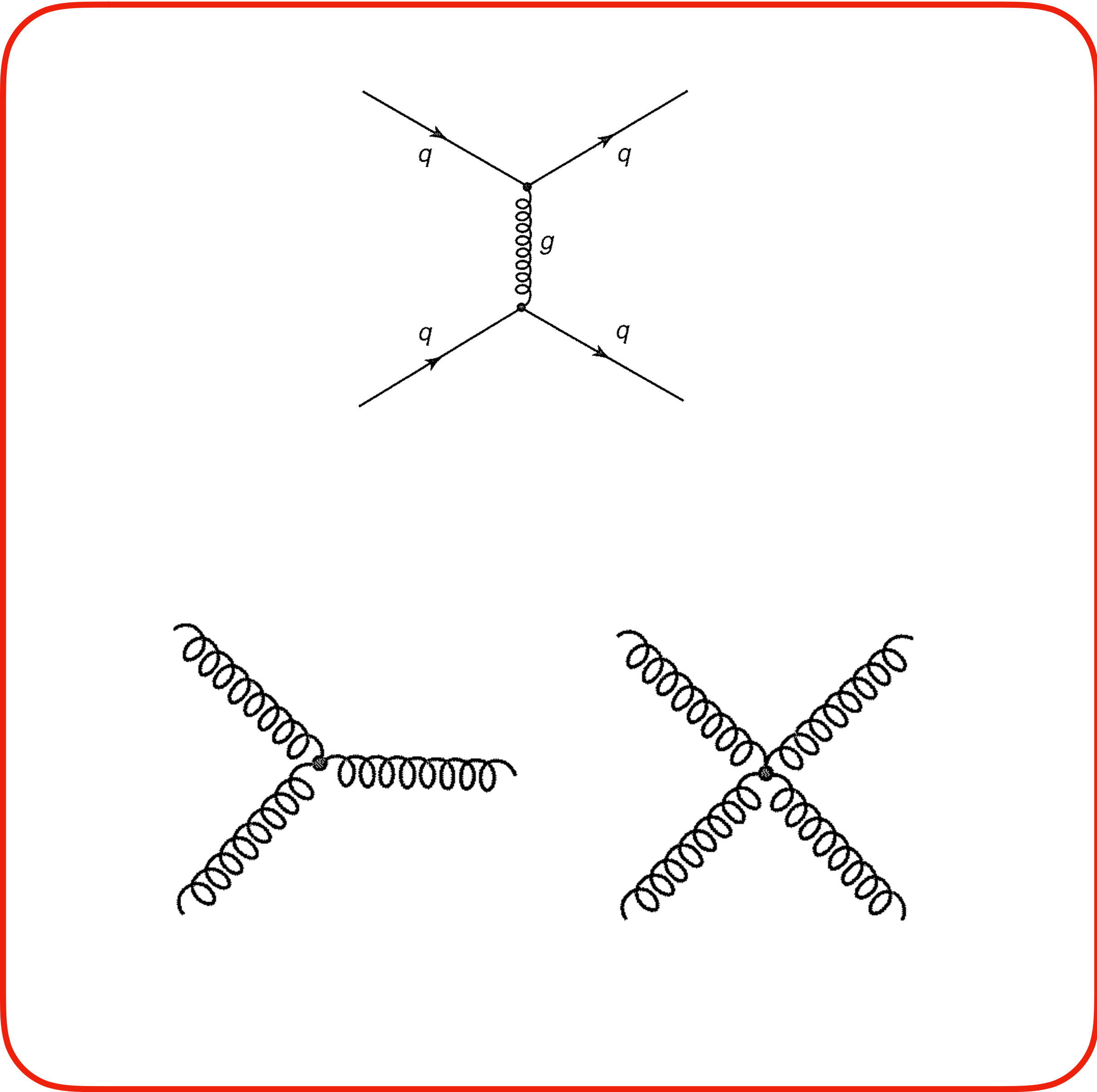
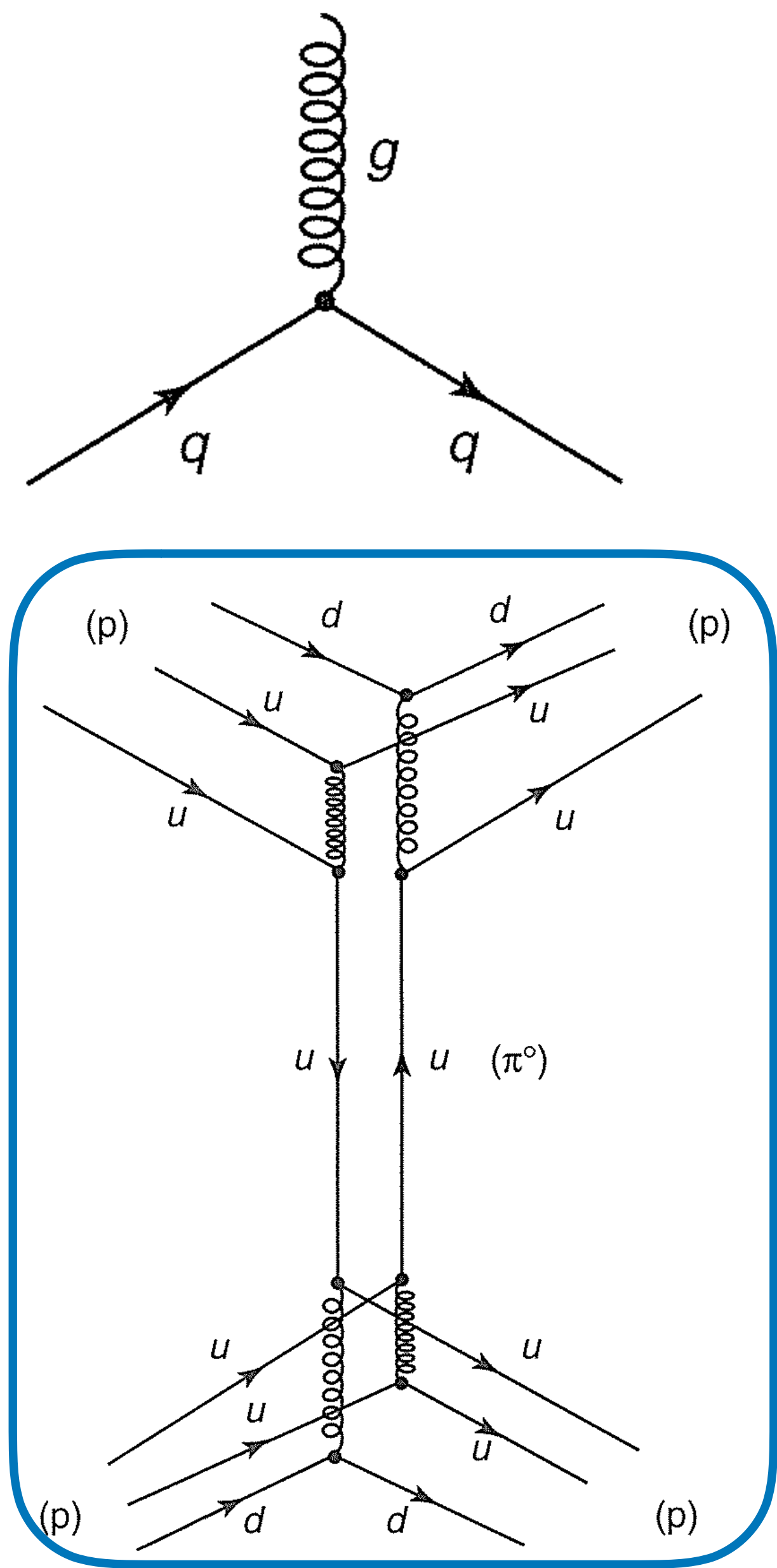
It's like a boiling lake.  
Creating virtual particles all the time

<https://www.youtube.com/watch?v=n0Nxu-Mqc-0>



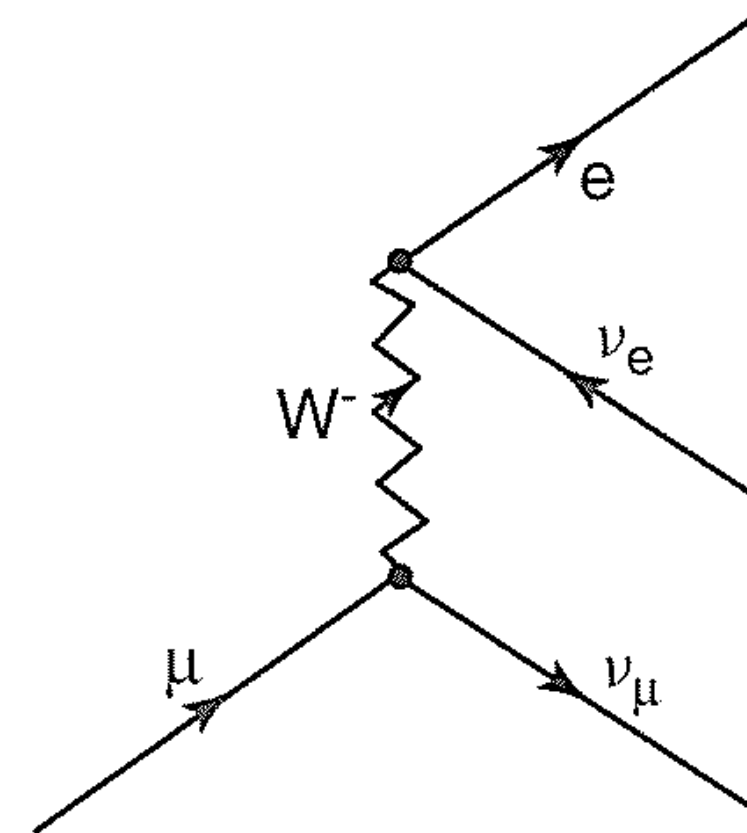
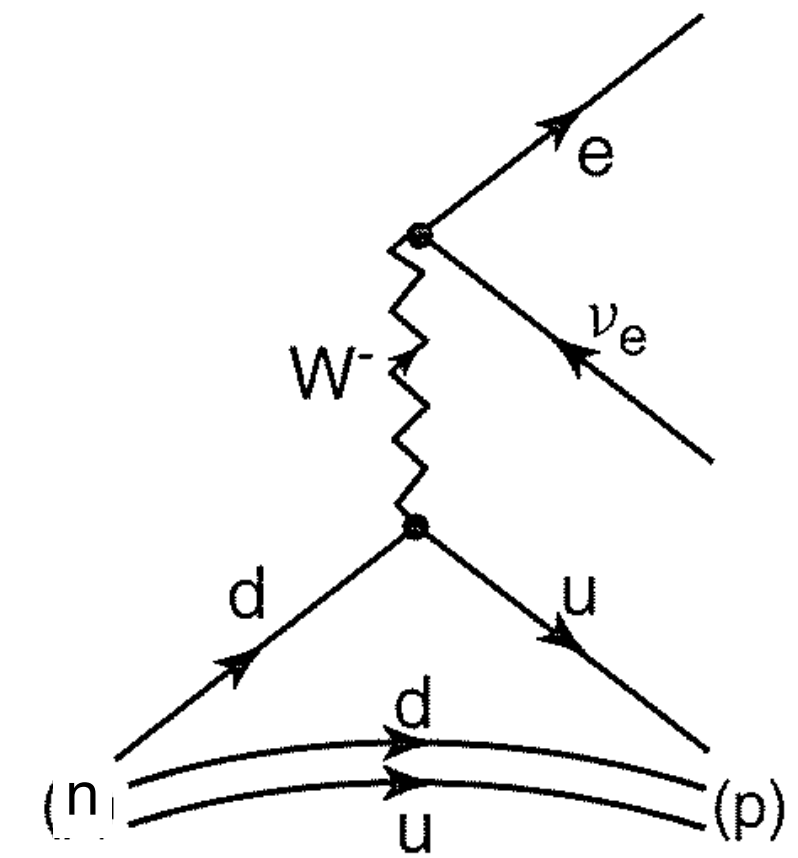
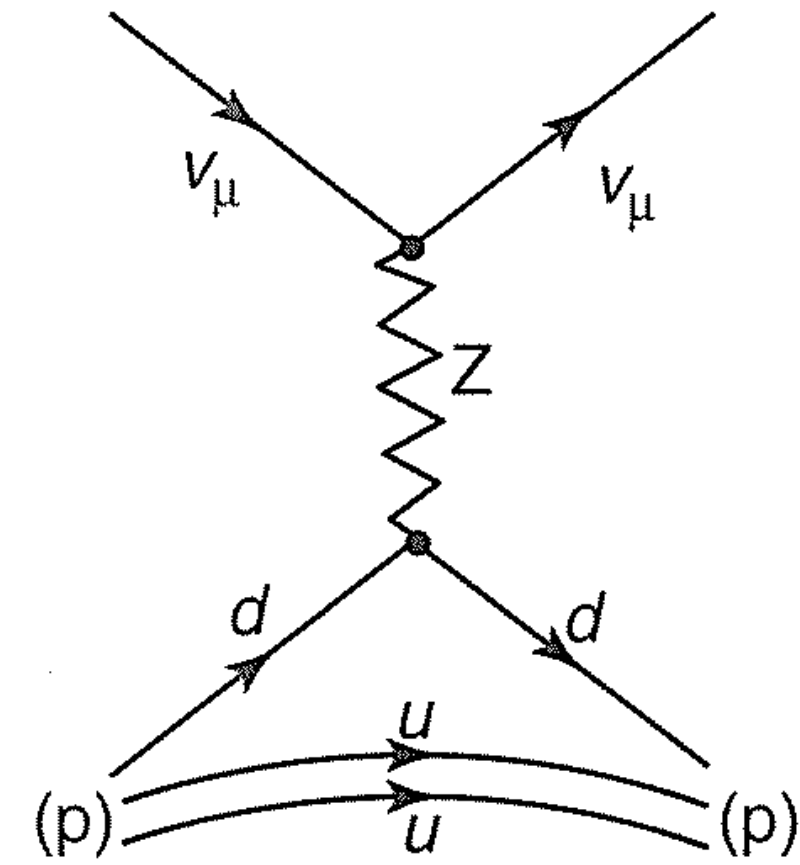
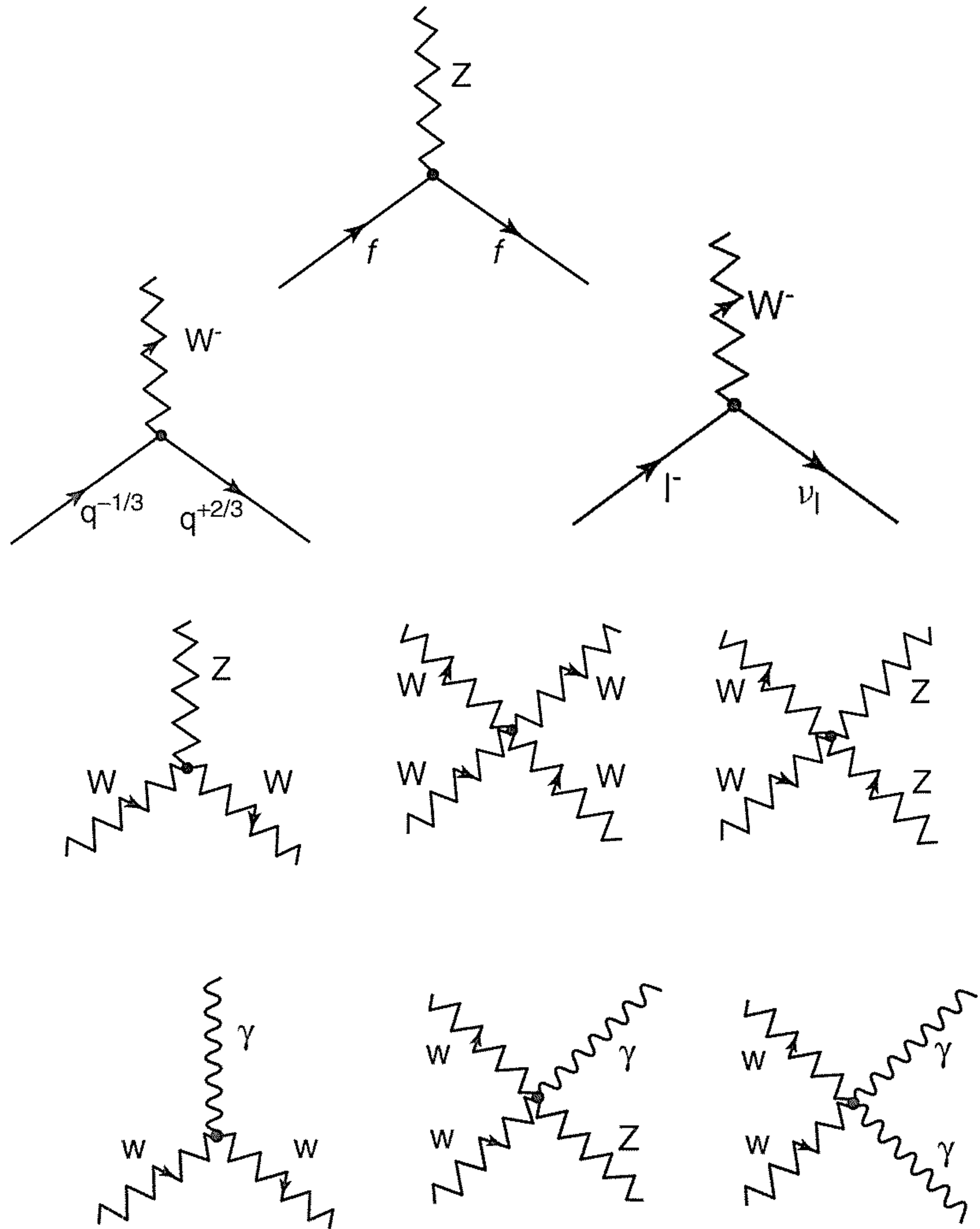


# Quantum Chromodynamics





# Weak Interactions



# So, what do we have?

*Charge:* All three interactions, of course, conserve electric charge. In the case of the weak interactions, the lepton (or quark) that comes out may not have the same charge as the one that went in, but if so, the difference is carried away by the  $W$ .

*Color:* The electromagnetic and weak interactions do not affect color. At a strong vertex the quark color does change, but the difference is carried off by the gluon. (The direct gluon–gluon couplings also conserve color.) However, since naturally occurring particles are always colorless, the observable manifestation of color conservation is pretty trivial: zero in, zero out.

*Baryon number:* In all the primitive vertices, if a quark goes *in*, a quark comes *out*, so the total number of quarks present is a constant. In this arithmetic we count *antiquarks* as *negative*, so that, for example, at the vertex  $q + \bar{q} \rightarrow g$  the quark number is zero before and zero after. Of course, we never see individual quarks, only baryons (with quark number 3), antibaryons (quark number  $-3$ ), and mesons (quark number zero). So, in practice, it is more convenient to speak

of the conservation of baryon number (1 for baryons,  $-1$  for antibaryons, and 0 for everything else). The baryon number is just  $\frac{1}{3}$  the quark number. Notice that there is no analogous conservation of *meson number*; since mesons carry zero quark number, a given collision or decay can produce as many mesons as it likes, consistent with conservation of energy.

*Lepton number:* The strong forces do not touch leptons at all; in an electromagnetic interaction the same particle comes out (accompanied by a photon) as went in; and in the weak interactions if a lepton goes in, a lepton comes out (not necessarily the *same* one, this time). So, lepton number is absolutely conserved. Until recently there appeared to be no cross-generation mixing among the leptons, so electron number, muon number, and tau number were all separately conserved. This remains true in most cases, but neutrino oscillations indicate that it is not absolute.



# So, what do we have?

*Flavor*: What about quark *flavor*? Flavor is conserved at a strong or electromagnetic vertex, but *not* at a weak vertex, where an up-quark may turn into a down quark or a strange quark, with nothing at all picking up the lost upness or supplying the ‘gained’ downness or strangeness. Because the weak forces *are* so weak, we say that the various flavors are *approximately* conserved. In fact, as you may remember, it was precisely this approximate conservation that led Gell-Mann to introduce the notion of strangeness in the first place. He ‘explained’ the fact that strange particles are always produced in pairs:

$$\pi^{-}(d\bar{u}) + p^{+}(uud) \rightarrow K^{+}(u\bar{s}) + \Sigma^{-}(dds)$$

for instance, but

$$\pi^{-}(d\bar{u}) + p^{+}(uud) \not\rightarrow \pi^{+}(u\bar{d}) + \Sigma^{-}(dds)$$

by arguing that the latter violates conservation of strangeness. (Actually, this *is* a possible *weak* interaction, but it will never be seen in the laboratory, because it must compete

against enormously more probable strong processes that do conserve strangeness.) In *decays*, however, the nonconservation of strangeness is very conspicuous, because for many particles this is the only way they *can* decay; there is no competition from strong or electromagnetic processes. The  $\Lambda$ , for instance, is the lightest strange baryon; if it is to decay, it must go to  $n$  (or  $p$ ) plus something. But the lightest strange meson is the  $K$ , and  $n$  (or  $p$ ) plus  $K$  weighs substantially more than the  $\Lambda$ . If the  $\Lambda$  decays at all (and it *does* as we know:  $\Lambda \rightarrow p^{+} + \pi^{-}$  64% of the time, and  $\Lambda \rightarrow n + \pi^{0}$  36% of the time), then strangeness cannot be conserved, and the reaction must proceed by the weak interaction. By contrast, the  $\Delta^{0}$  (with a strangeness of zero) can go to  $p^{+} + \pi^{-}$  or  $n + \pi^{0}$  by the *strong* interaction, and its lifetime is accordingly much shorter.

# Which of these are allowed?

## Home work

(a)  $p + \bar{p} \rightarrow \pi^+ + \pi^0$

(c)  $\Sigma^0 \rightarrow \Lambda + \pi^0$

(e)  $e^+ + e^- \rightarrow \mu^+ + \mu^-$

(g)  $\Delta^+ \rightarrow p + \pi^0$

(i)  $e + p \rightarrow \nu_e + \pi^0$

(k)  $p \rightarrow e^+ + \gamma$

(m)  $n + \bar{n} \rightarrow \pi^+ + \pi^- + \pi^0$

(o)  $K^- \rightarrow \pi^- + \pi^0$

(q)  $\Sigma^0 \rightarrow \Lambda + \gamma$

(s)  $\Xi^0 \rightarrow p + \pi^-$

(u)  $\pi^0 \rightarrow \gamma + \gamma$

(b)  $\eta \rightarrow \gamma + \gamma$

(d)  $\Sigma^- \rightarrow n + \pi^-$

(f)  $\mu^- \rightarrow e^- + \bar{\nu}_e$

(h)  $\bar{\nu}_e + p \rightarrow n + e^+$

(j)  $p + p \rightarrow \Sigma^+ + n + K^0 + \pi^+ + \pi^0$

(l)  $p + p \rightarrow p + p + p + \bar{p}$

(n)  $\pi^+ + n \rightarrow \pi^- + p$

(p)  $\Sigma^+ + n \rightarrow \Sigma^- + p$

(r)  $\Xi^- \rightarrow \Lambda + \pi^-$

(t)  $\pi^- + p \rightarrow \Lambda + K^0$

(v)  $\Sigma^- \rightarrow n + e + \bar{\nu}_e$